Asp.net

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Agenda

- Introduction
- Evolution /Need
- .Net Framework
- Concept ASP.NET/Life Cycle
- Working of ASP.NET
- Visual studio and ASP.NET
- MODELS

Agenda

- Web Pages
- Web Forms
- Advantages
- Disadvantages
- Controls
- Validation Controls
- Data binding
- Regular Expression
- Exception Handling

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introduction

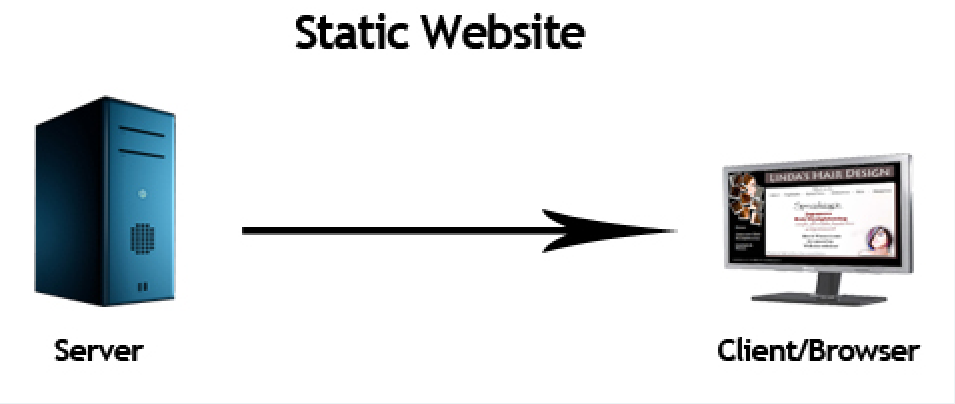
1. Need to develop languages such and programming tools that could integrate with the web.

- Separate tiny applications that are executed by server side calls – CGI (Common gateway interface)
- Scripts that are interpreted by a server side resource : Classic ASP(Active server pages.)
- **Problems in ASP**
  - No IDE for developers with classic ASP.
  - ASP code is interpreted.
  - Length Code.

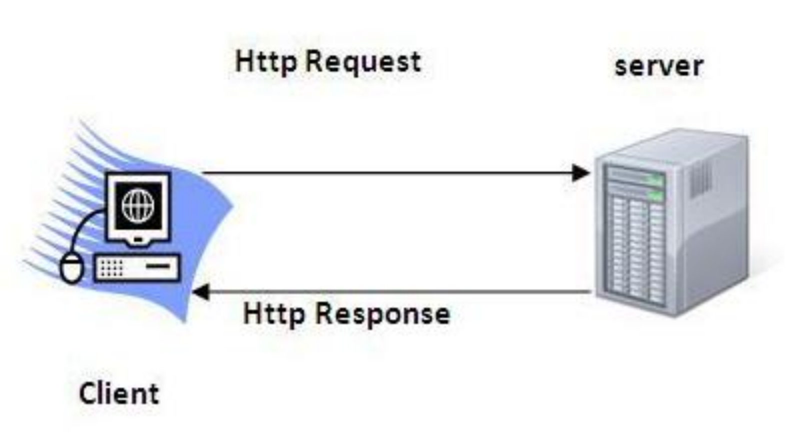
Website/Applications

- Website is a collection of related web pages that may contain text, images, audio and video. The first page of a website is called home page. Each website has specific internet address (URL) that you need to enter in your browser to access a website.
- Website is hosted on one or more servers and can be accessed by visiting its homepage using a computer network. A website is managed by its owner that can be an individual, company or an organization.

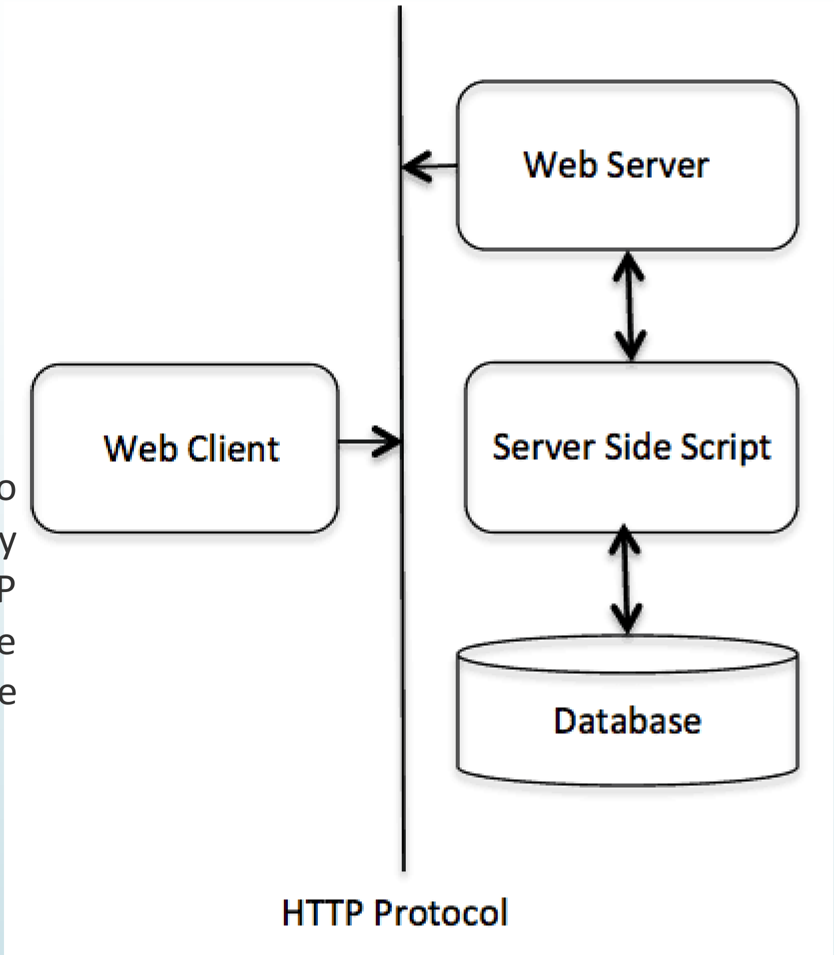
introduction



Protocol: HTTP



HTTP request: The request sent by the computer to a web server, contains all sorts of potentially interesting information; it is known as HTTP requests.The HTTP client sends the request to the server in the form of request message like GET,POST methods



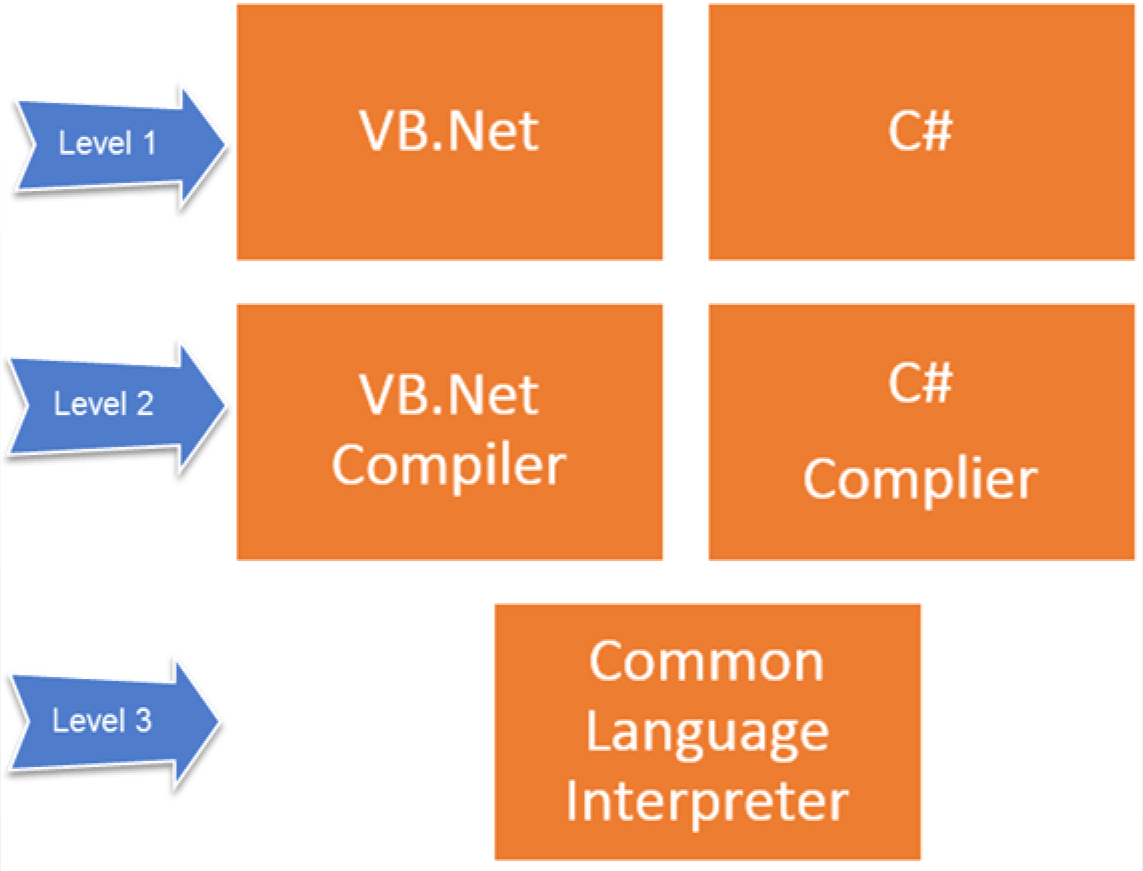
.NET framework

- **.Net Framework Architecture** is a programming model for the .Net platform that provides an execution environment and integration with various programming languages for simple development and deployment of various Windows and desktop applications. It consists of class libraries and reusable components.

- .NET is a developer platform made up of tools, programming languages, and libraries for building many different types of applications
- It is a software devlopment framework from mocrosoft.It provide a controlled programming environment where software can be devloped,installed and executed on window based operating systems.
- The .net languages includes Visual basic,c#,jscript,.net(a server side version)etc.

- There are various implementations of .NET. Each implementation allows .NET code to execute in different places—Linux, macOS, Windows, iOS, Android, and many more.

.NET framework



Components: **Class libraries(**provides a set of APIs and types for common functionality **)+ CLR**

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.NET framework

- .NET applications are written in the C#, F#, or Visual Basic programming language. Code is compiled into a language-agnostic Common Intermediate Language **(CIL**). Compiled code is stored in assemblies—files with a .dll or .exe file extension.

When an app runs, the **CLR** takes the assembly and uses a just-in-time compiler (JIT) to turn it into machine code that can execute on the specific architecture of the computer it is running on.

- **CLR**: A .NET Common Language Runtime-compliant compiler compiles a .NET web page into an intermediate language, then a JIT compiler converts the intermediate code to native machine code, and this machine code is eventually run on the processor.
- ASP.net applications are executed via a sequence of HTTP requests and HTTP responses
- In ASP.NET
  - Client web browser request ASPX pages.
  - I IS passes the request to the ASP.NET engine on the server.
  - The web server executes the ASPX pages and produce XHTML +CSS+ java script .
  - ASP.net file is returned to the browser as plain HTML.

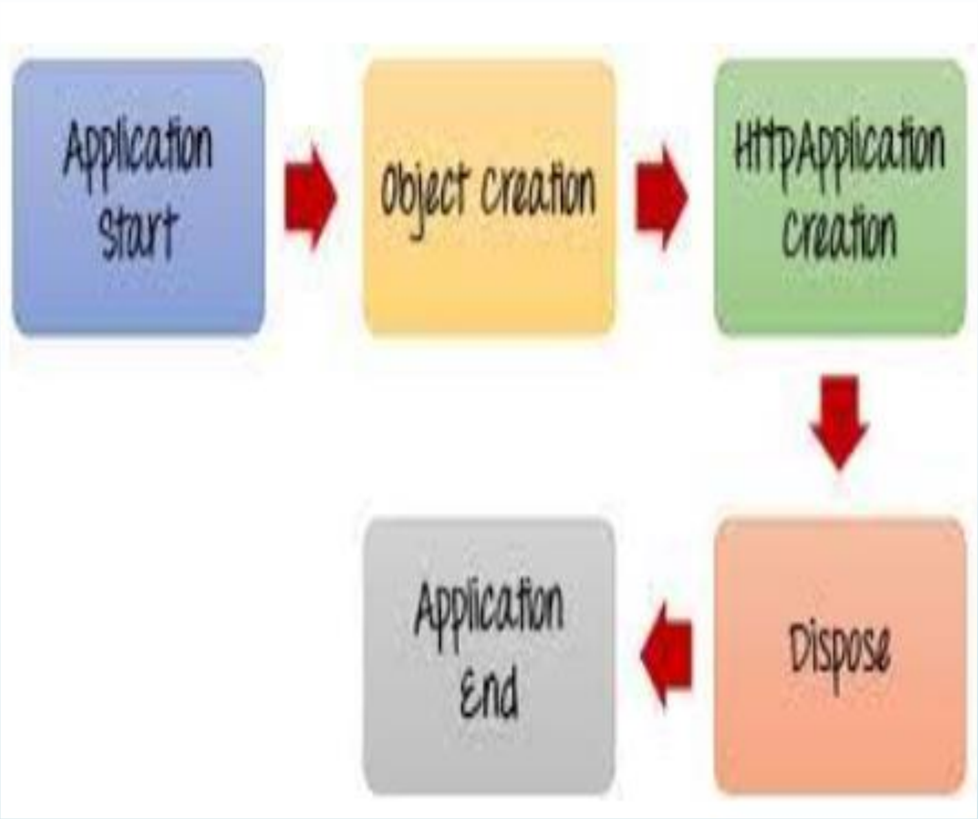
Working of ASP.NET

- ASP.NET is server side technologies. It enable computer code to be executed by an Internet server. **When a browser requests an ASP or ASP.NET file, the ASP engine reads the file, executes any code in the file, and returns the result to the browser**.

- **An open-source web application framework developed by Microsoft** in web development design is ASP.Net. The application is developed on the server-side and helps in making dynamic web pages.

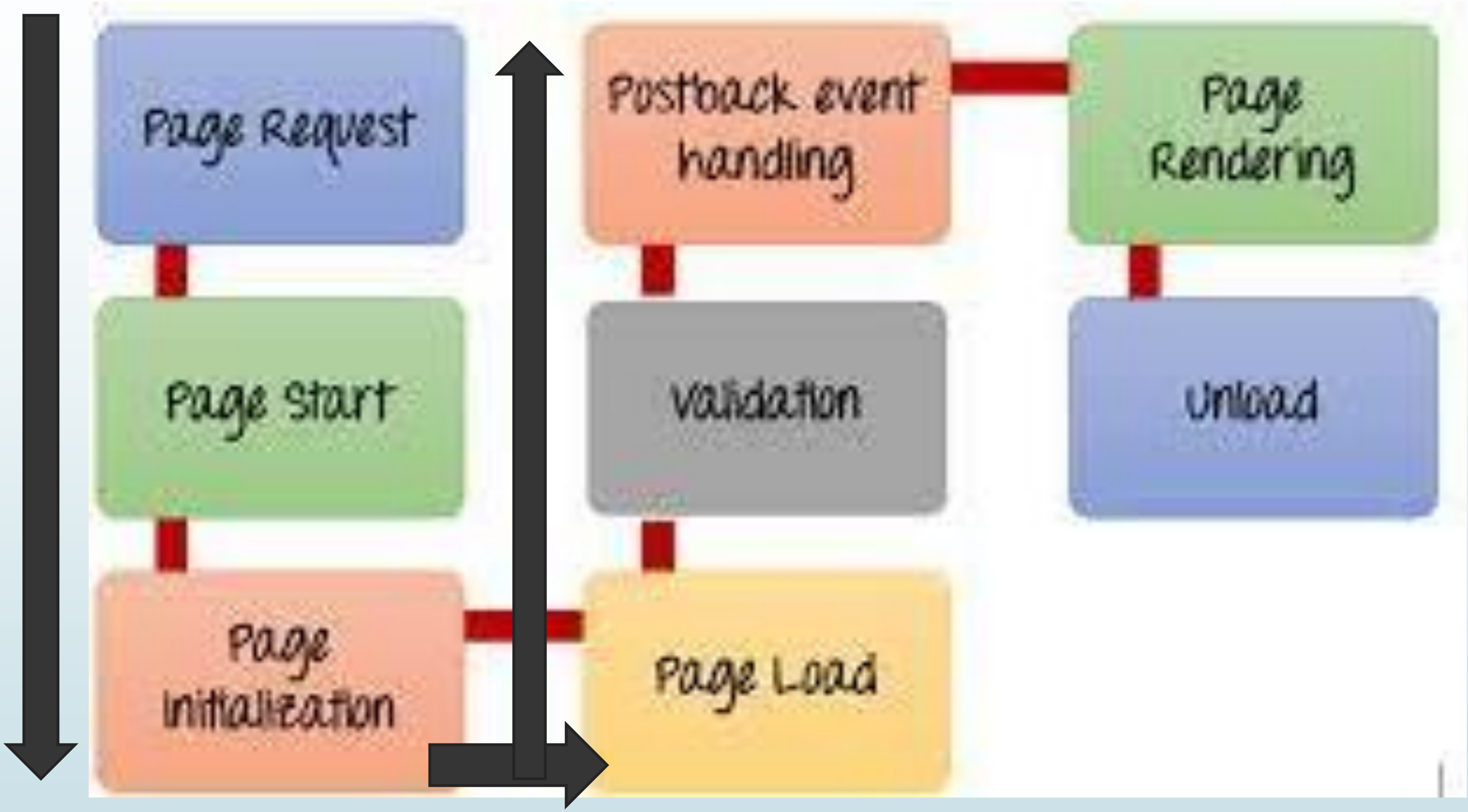
- **ASP.Net is written in .Net languages and has an Apache license**. It operates in Windows, Linux, and MacOS. It uses HTTP commands and works with HTTP protocol to have proper communication of browser and server level. It simplifies the working of websites and applications as it builds a library of many codes and tools. We can use ASP.Net for front-end and back-end development.

ASP.net Life cycle



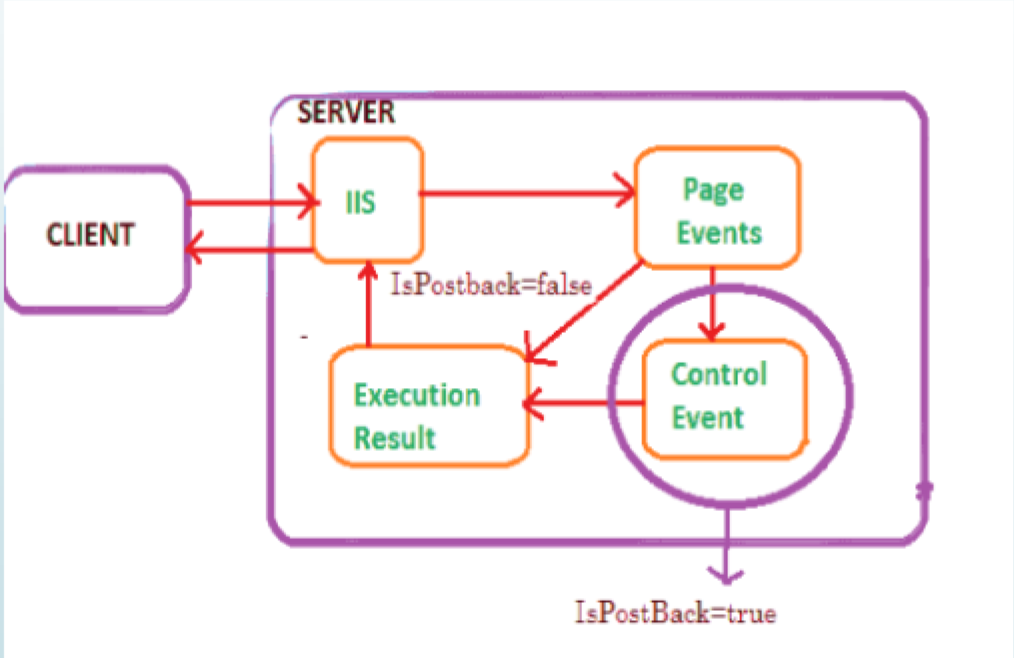
- **1) Application Start** – The life cycle of an ASP.NET application starts when a request is made by a user. This request is to the Web server for the ASP.Net Application. This happens when the first user normally goes to the home page for the application for the first time. During this time, there is a method called Application_start which is executed by the web server. Usually, in this method, all global variables are set to their default values.
- **Object creation** – The next stage is the creation of the HttpContext, HttpRequest & HttpResponse by the web server. The HttpContext is just the container for the HttpRequest and HttpResponse objects. The HttpRequest object contains information about the current request, including cookies and browser information. The HttpResponse object contains the response that is sent to the client.
- **HttpApplication creation** – This object is created by the web server. It is this object that is used to process each subsequent request sent to the application.
- **Dispose** – This event is called before the application instance is destroyed.
- **Application End** – This is the final part of the application. In this part, the application is finally unloaded from memory.

Life cycle of Asp.Net page



Page life cycle events:Post Back

Postback is actually sending all the information from client to web server, then web server process all those contents and returns back to the client. Most of the time ASP control will cause a post back (e. g. buttonclick) but some don't unless you tell them to do In certain events (Listbox Index Changed,RadioButton Checked etc..) in an ASP.NET page upon which a PostBack might be needed.



Controls:

- ASP.NET uses five types of web controls, which are:
- HTML controls
- HTML Server controls
- ASP.NET Server controls
- ASP.NET Ajax Server controls
- User controls and custom controls

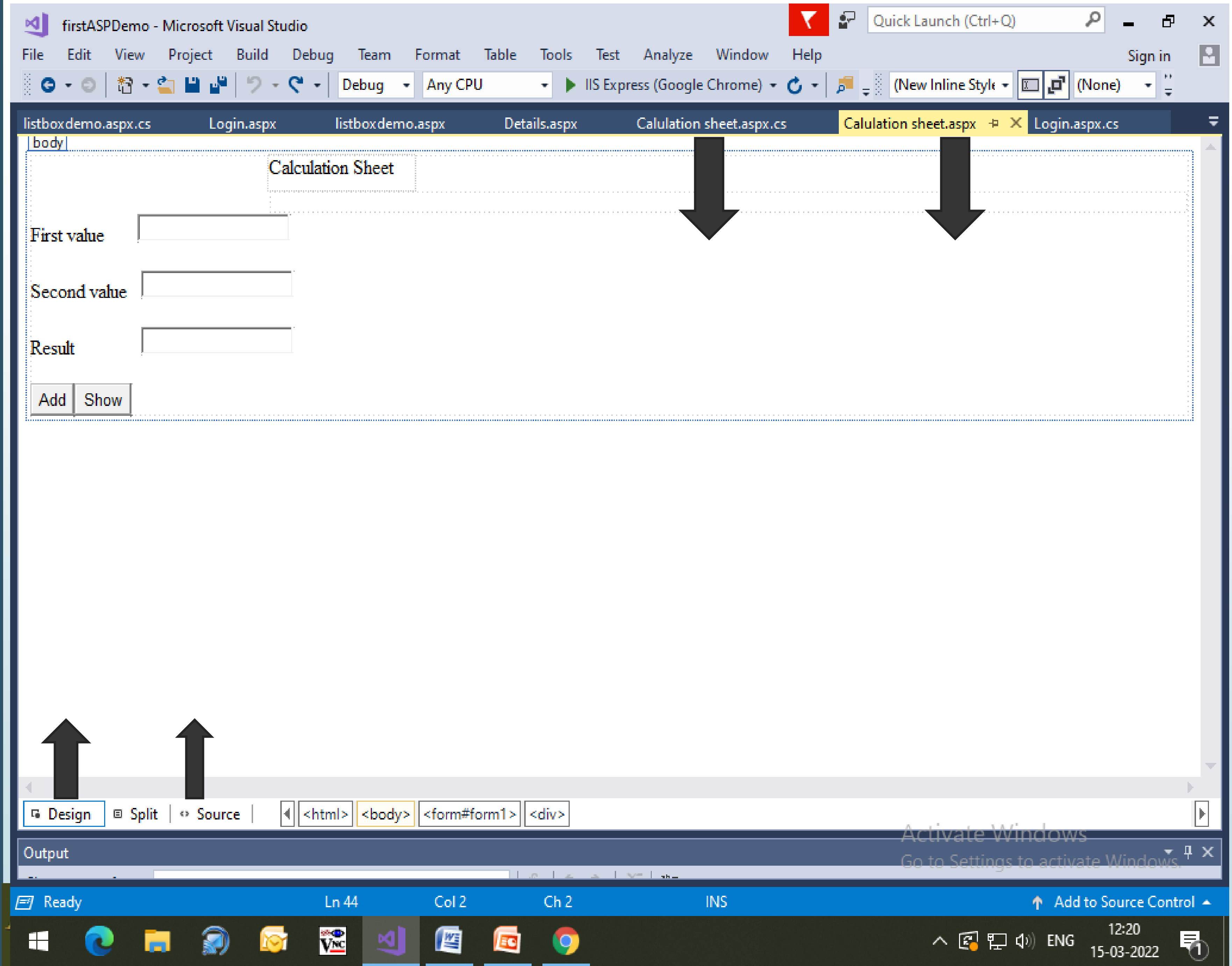
Server controls:

ASP.NET server controls are the primary controls used in ASP.NET. These controls can be grouped into the following categories:

- **Validation controls** - These are used to validate user input and they work by running client-side script.
- **Data source controls** - These controls provides data binding to different data sources.
- **Data view controls** - These are various lists and tables, which can bind to data from data sources for displaying.
- **Personalization controls** - These are used for personalization of a page according to the user preferences, based on user information.
- **Login and security controls** - These controls provide user authentication.
- **Master pages** - These controls provide consistent layout and interface throughout the application.
- **Navigation controls** - These controls help in navigation. For example, menus, tree view etc.
- **Rich controls** - These controls implement special features. For example, AdRotator, FileUpload, and Calender

Server Controls

- The syntax for using server controls is:
- <asp:controlType ID ="ControlID" runat="server" Property1=value1 [Property2=value2] />
- In addition, visual studio has the following features, to help produce in error-free coding:
- Dragging and dropping of controls in design view
- IntelliSense feature that displays and auto-completes the properties
- The properties window to set the property values directly


firstASPDemo - Microsoft Visual Studio
Quick Launch (Ctrl+Q)
File Edit View Project Build Debug Team Format Table Tools Test Analyze Window Help
Sign in
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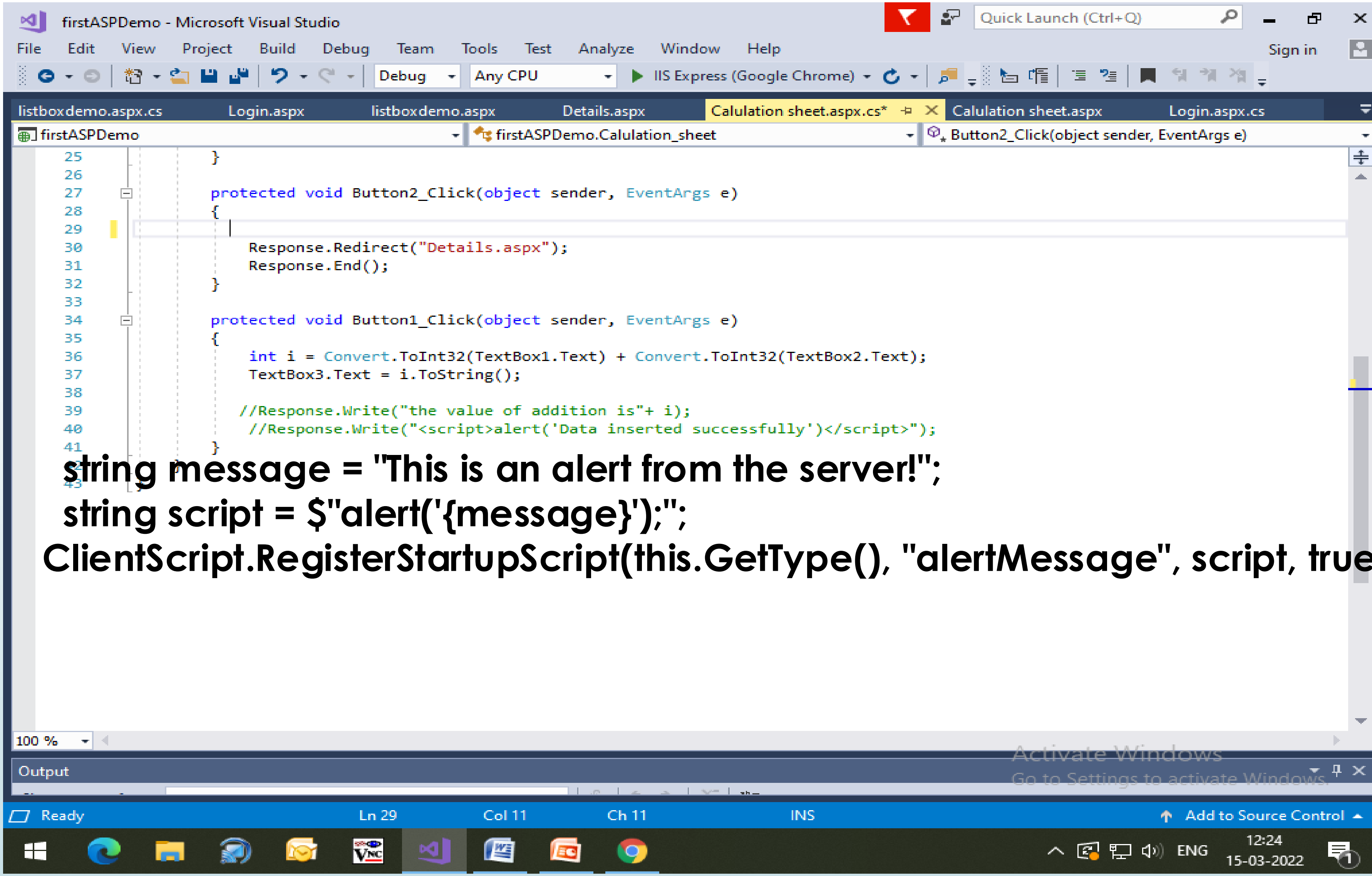
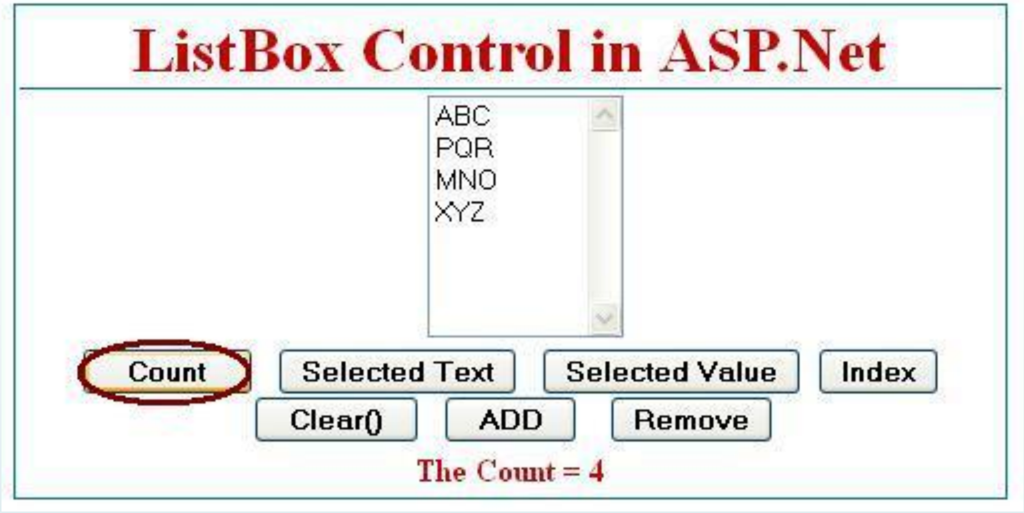



```
string message = "This is an alert from the server!";
string script = $"alert('{message}');";
ClientScript.RegisterStartupScript(this.GetType(), "alertMessage", script, true);
```

Controls



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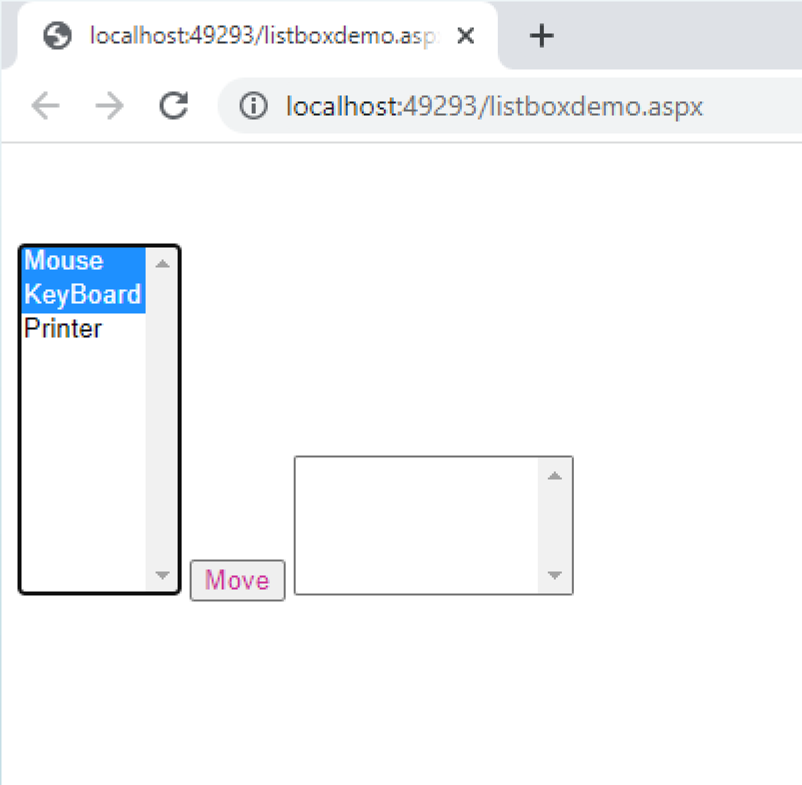
List Box

- There are other some important properties of listbox :
- **ListBox1.Items.Count** = Return the total count of items in the listbox.
- **ListBox1.Items.Add("ItemName")** = Add the new item in to the listbox control.
- **ListBox1.Items.Insert(int index, "ItemName")** = Add the new item at specific position in listbox control.
- **ListBox1.Items.Remove("ItemName")** = Remove the item from the listbox control.
- **ListBox1.Items.RemoveAt(int index)** = Remove the item from the listbox at desired position index.
- **ListBox1.Items.Clear()** = Clear all the items from listbox control.
- **ListBox1.SelectedItem.Text** = Returns the Text value of selected item of the listbox.
- **ListBox1.SelectedValue** = Returns the Value property of the selected item of the listbox.
- **ListBox1.SelectedIndex** = Returns the Index of selected item of listbox. (Index always start from Zero).
- Here, is the code for bind listbox control from database.
- **ListBox1.DataSource** = DataTable OR DataSet
- **ListBox1.DataTextField** = Bind the Text to listbox (It is visible in listbox)
- **ListBox1.DataValueField** = Bind the Value to listbox (It is Invisible in listbox)
- **ListBox1.DataBind();**

- protected void btncount_Click(object sender, EventArgs e)
- { Label1.Text **= "The Count = "+ListBox1.Items.Count.ToString();** }
- string text = **listBox1.GetItemText(listBox1.SelectedItem)**
- foreach (var item in listBox1.SelectedItems)

```
{
list1 += listBoxFiles.GetItemText(item) + "\n";
}
```



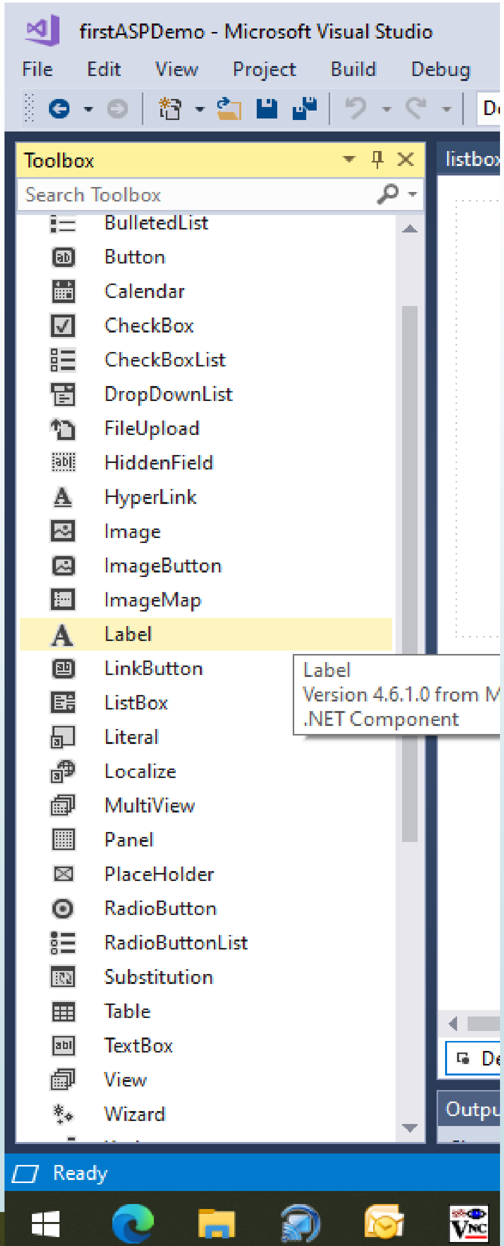
```
namespace firstASPDemo
{
    public partial class Shop : System.Web.UI.Page
    {
        ArrayList arraylist1 = new ArrayList();
        protected void Page_Load(object sender, EventArgs e)
        {
            this.UnobtrusiveValidationMode = System.Web.UI.UnobtrusiveValidationMode.None;
            ListBox1.Items.Add("Mouse");
            ListBox1.Items.Add("KeyBoard");
            ListBox1.Items.Add("Printer"); //ListBox1.Items.Add
```

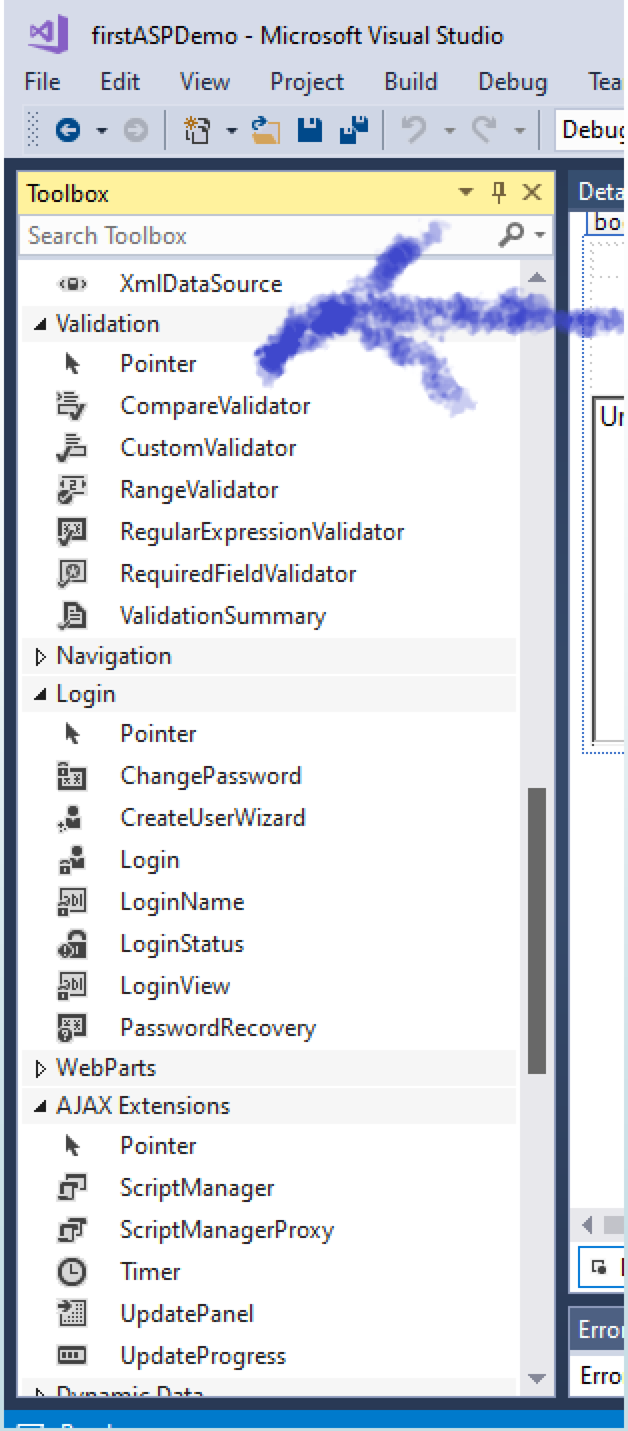
```
        protected void Button1_Click(object sender, EventArgs e)
        {    //to copy one item
             //foreach (var item in (ListBox1.SelectedItem.ToString()))
             //{
             //    ListBox2.Items.Add(item.ToString());}

            if (ListBox1.SelectedIndex >= 0)
            {
                for (int i = 0; i < ListBox1.Items.Count; i++)
                {
                    if (ListBox1.Items[i].Selected)
                    {
                        if (!arraylist1.Contains(ListBox1.Items[i]))
                        {
                            arraylist1.Add(ListBox1.Items[i]);

                        }
                    }
                }
            }
            for (int i = 0; i < arraylist1.Count; i++)
            {
                if (!ListBox2.Items.Contains(((ListItem)arraylist1[i])))
                {
                    ListBox2.Items.Add(((ListItem)arraylist1[i]));
                }
                ListBox1.Items.Remove(((ListItem)arraylist1[i]));
            }
            ListBox2.SelectedIndex = -1;
        }
```

Validation Controls





Validation Controls

- **Validation is of two types**
- Client Side
- Server Side

You can use ASP.NET validation, which will ensure client, and server validation. It work on both end; first it will work on client validation and than on server validation. At any cost server validation will work always whether client validation is executed or not. So you have a safety of validation check.

For client script .NET used JavaScript. WebUIValidation.js file is used for client validation by .NET

- There are six types of validation controls in ASP.NET
    1. RequiredFieldValidation Control
    2. CompareValidator Control
    3. RangeValidator Control
    4. RegularExpressionValidator Control
    5. CustomValidator Control
    6. ValidationSummary
- **Important points for validation controls**
    - ControlToValidate property is mandatory to all validate controls.
    - One validation control will validate only one input control but multiple validate control can be assigned to a input control.
    1. [Visit Validation Controls]

```
//resolve error
protected void Page_Load(object sender, EventArgs e)//in all forms of application
    {
        this.UnobtrusiveValidationMode = System.Web.UI.UnobtrusiveValidationMode.None;
    }
```

and display=dynamic

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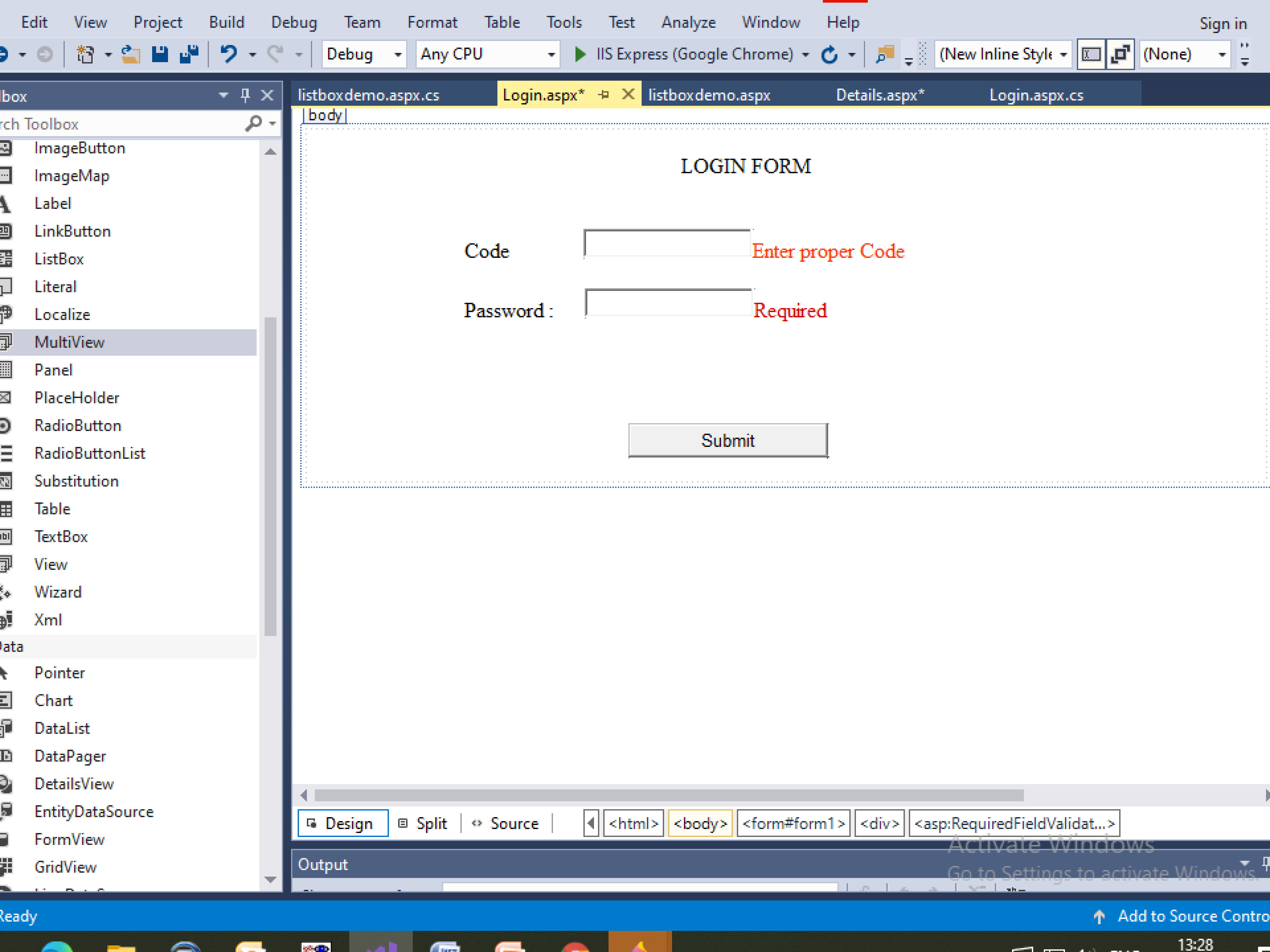
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<form#form1>
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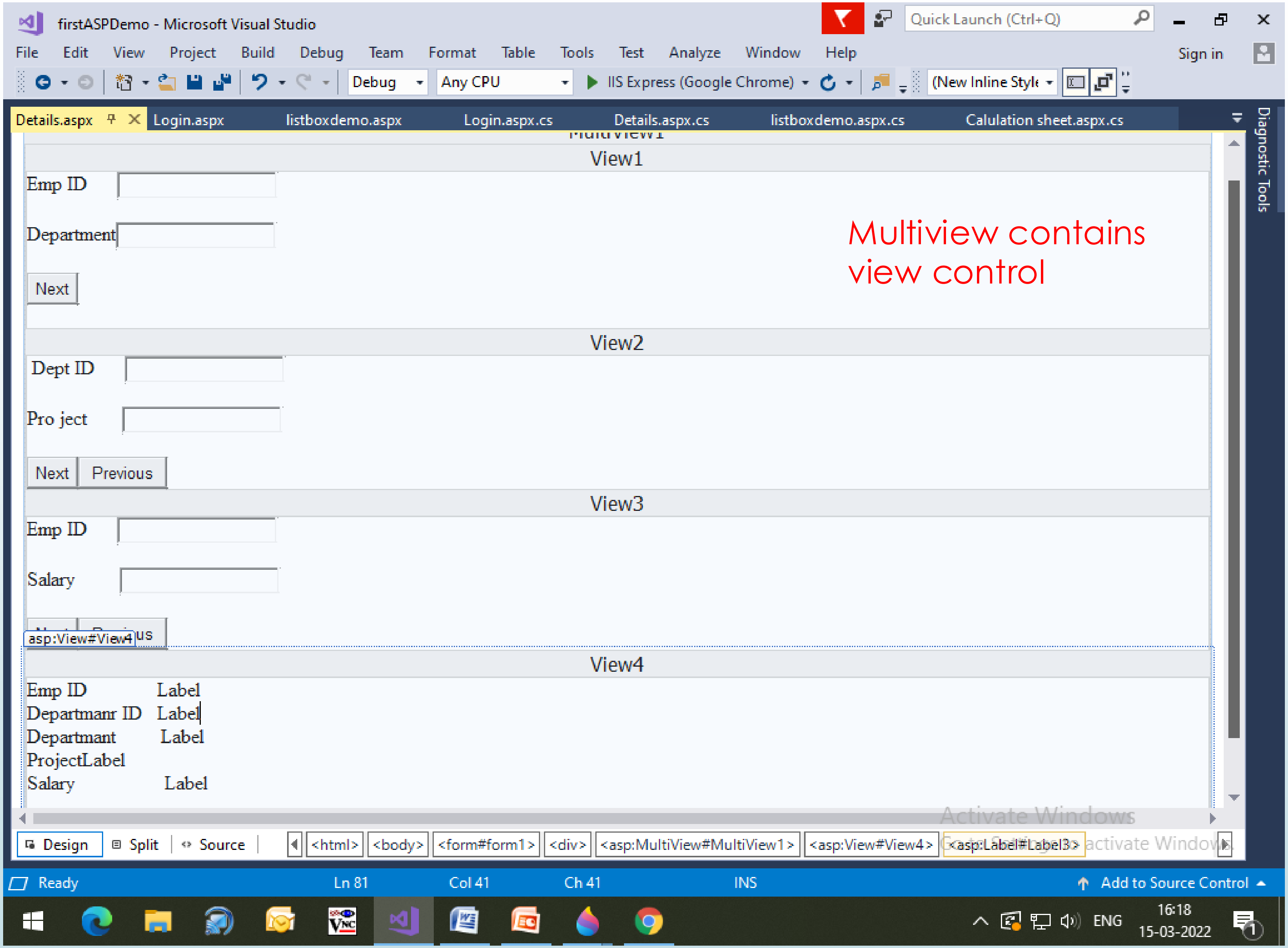
13:28

```
                    
            <br />
            <br />
            Code              <asp:TextBox ID="TextBox

            <asp:RangeValidator ID="RangeValidator1" runat="server" ControlToValidate="TextBox1"
                ErrorMessage="Enter proper Code" MaximumValue="2" MinimumValue="10"
                Display="Dynamic" ForeColor="#FF3300"
                ValidationGroup="one"></asp:RangeValidator>
            <br />
            <br />

            Password :     
            <asp:TextBox ID="TextBox2" runat="server" OnTextChanged="TextBox2_TextChanged" style="margin-left: 0px"
                CausesValidation="True"></asp:TextBox>
            <asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server"
                ControlToValidate="TextBox2"
                ErrorMessage="Required"
                Display="Dynamic" ForeColor="#CC0000"
                InitialValue="Enter password" ValidationGroup="one">

            </asp:RequiredFieldValidator>
```

Data Grid/Data List

Multiview and View control



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Tree View Control:Stored Procedure

TreeVIew Control:Navigation Controls

Rich Controls:AdRotator:

The AdRotator is one of the rich web server control of asp.net. AdRotator control is used to display a sequence of advertisement images as per given priority of image.

Follow below step to create Adrotator control in asp.net web application.

Step 1 – Open Visual Studio –> Create a new empty web application / website.

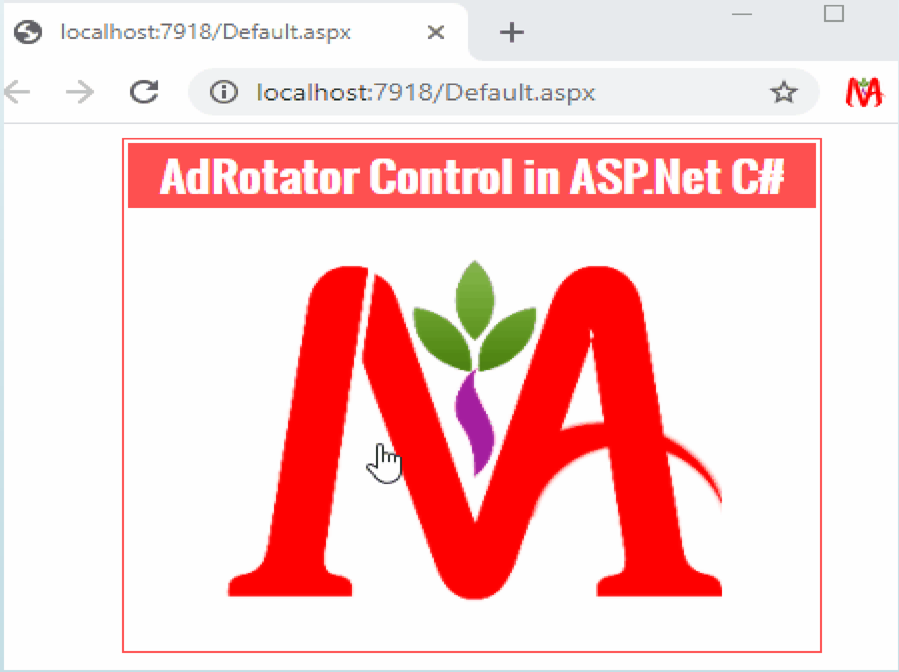
Step 2 – Create New web page for display AdRotator control.

step 3 – Drag and drop AdRotator Control on web page from toolbox.

step 4 – Right click on Solution Explorer –> **Add New Item** –> Add New **XML File** in project for write advertisement detail.

step 5 – Write code in xml file for advertisement.

step 6 – Assign XML File to **AdvertisementFile** Property of AdRotator control.



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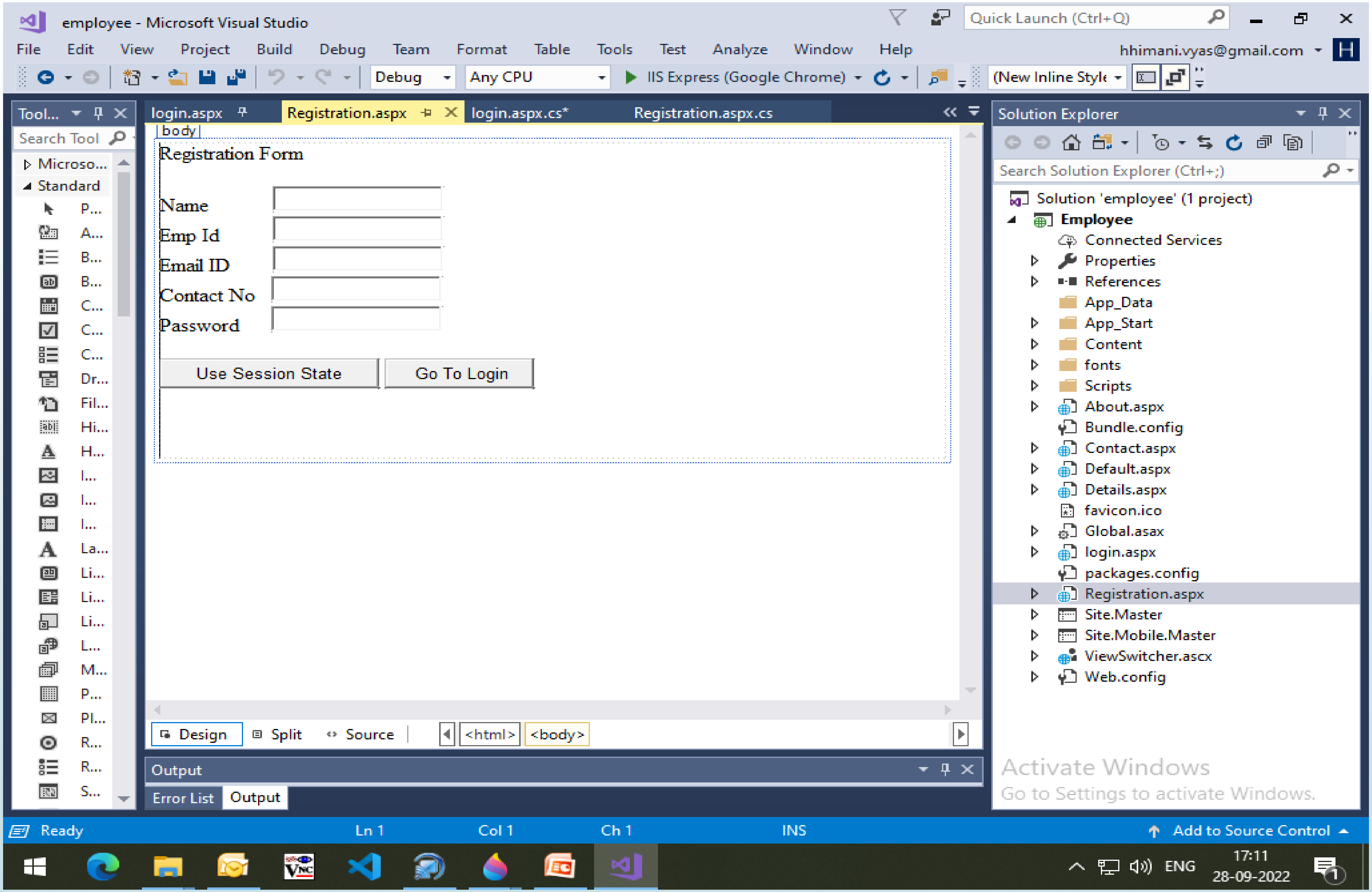
Sessions and Cookies:employees.sln



ogin.aspx | login.aspx.cs* | Registration.aspx.cs | Details.aspx | login.aspx.designer.cs | Details

Employee | Employee.Registration | TextBox5

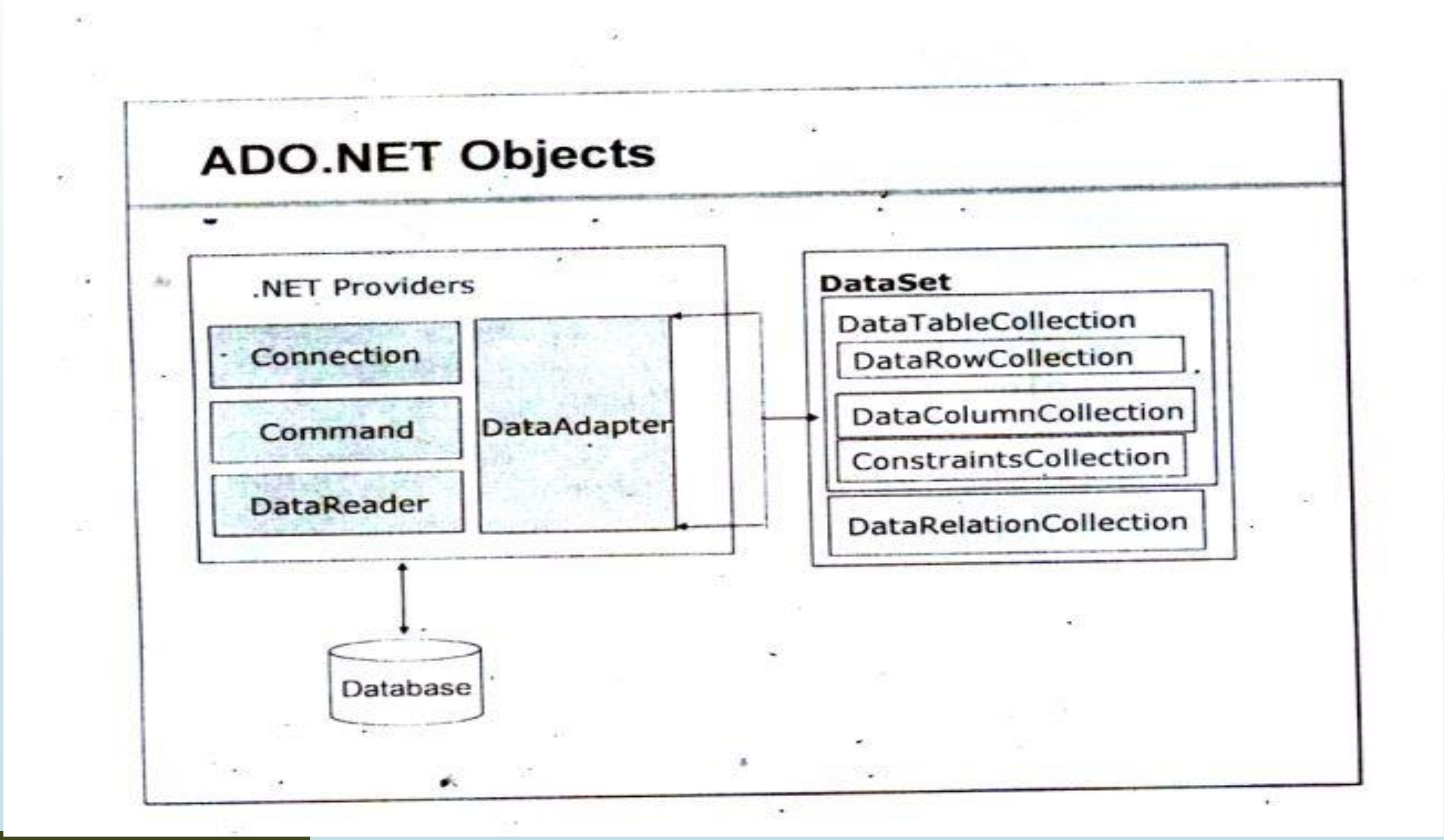
```
public partial class Registration : System.Web.UI.Page
{
    protected void Page_Load(object sender, EventArgs e)
    {

    }

    protected void Button1_Click(object sender, EventArgs e)
    {


        Response.Write("<script>alert('mail id is :" + TextBox2.Text + "');</script>");
        Session["mailid"] = TextBox2.Text;
        // ***use of response.redirect
        //Response.Redirect("login.aspx?id" + TextBox2.Text);
        //****use of cookie
        HttpCookie cID = new HttpCookie("cID");
        cID.Value = TextBox2.Text;
        Response.Cookies.Ad
        Response.Write("<sc                                        " + TextBox2.Text + "');</sc
        cID.Expires = DateT

        HttpCookie visit = new HttpCookie("visit");
        visit.Value = DateTime.Now.ToString();
        Response.Write("<script>alert('Time of visit :" + visit.Value + "');</script>");
    }
```

(field) TextBox Registration.TextBox2
TextBox2 control.

ADO.Net

- ADO is a rich set of classes, interfaces, structures, and enumerated types that manage data access from various types of data stores.
- Enterprise applications handle a large amount of data. This data is primarily stored in relational databases, like Oracle, SQL Server, Access, and so on. These databases use Structured Query Language (SQL) for retrieval of data.
- To access enterprise data from a .NET application, an interface was needed. This interface acts as a bridge between an RDBMS system and a .Net application. ADO.NET is such an interface that is created to connect .NET applications to RDBMS systems.
- Open Database Connectivity (ODBC)
- Data Access Objects (DAO)
- Remote Data Objects (RDO)
- Active X Data Objects (ADO)

- Refrence:https://www.c-sharpcorner.com/UploadFile/d0a1c8/database-programming-with-ado-net/



Data base Connectivity:Connection

- **Connection** – To work with the data in a database, the first obvious step is the connection. The connection to a database normally consists of the below-mentioned parameters.
  - **Database name or Data Source** – The first important parameter is the database name. Each connection can only work with one database at a time.
  - **Credentials** – The next important aspect is the 'username' and 'password'. This is used to establish a connection to the database.
  - **Optional parameters** – You can specify optional parameters on how .net should handle the connection to the database. For example, one can specify a parameter for how long the connection should stay active.
- **Selecting data from the database** – Once the connection is established, data is fetched from the database. ASP.Net has the ability to execute 'sql' select command against the database. The 'sql' statement can be used to fetch data from a specific table in the database.
- **Inserting data into the database** – ASP.Net is used to insert records into the database. Values for each row that needs to be inserted in the database are specified in ASP.Net.
- **Updating data into the database** – ASP.Net can also be used to update existing records into the database. New values can be specified in ASP.Net for each row that needs to be updated into the database.
- **Deleting data from a database** – ASP.Net can also be used to delete records from the database. The code is written to delete a particular row from the database.

Command object

A Command object executes SQL statements on the database. These SQL statements can be SELECT, INSERT, UPDATE, or DELETE. It uses a connection object to perform these actions on the database.

A Connection object specifies the type of interaction to perform with the database, like SELECT, INSERT, UPDATE, or DELETE.

A Command object is used to perform various types of operations, like SELECT, INSERT, UPDATE, or DELETE on the database.

SELECT

```
cmd =new SqlCommand("select * from Employee", con);
```

INSERT

```
cmd = new SqlCommand("INSERT INTO Employee(Emp_ID,Emp_Name)VALUES ('" + aa + "','" + bb + "')", con);
```

UPDATE

```
SqlCommand cmd =new SqlCommand("UPDATE Employee SET

Emp_ID ='" + aa + "', Emp_Name ='" + bb + "' WHERE Emp_ID = '" + aa + "'", con);
```

DELETE

```
cmd =new SqlCommand("DELETE FROM Employee where Emp_ID='" + aa + "'", con);
```

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A Command object exposes several execute methods like:

ExecuteScaler()

Executes the query, and returns the first column of the first row in the result set returned by the query. Extra columns or rows are ignored.

ExecuteReader()

Display all columns and all rows in the client-side environment.

In other words, we can say that they display datatables client-side.

ExecuteNonQuery()

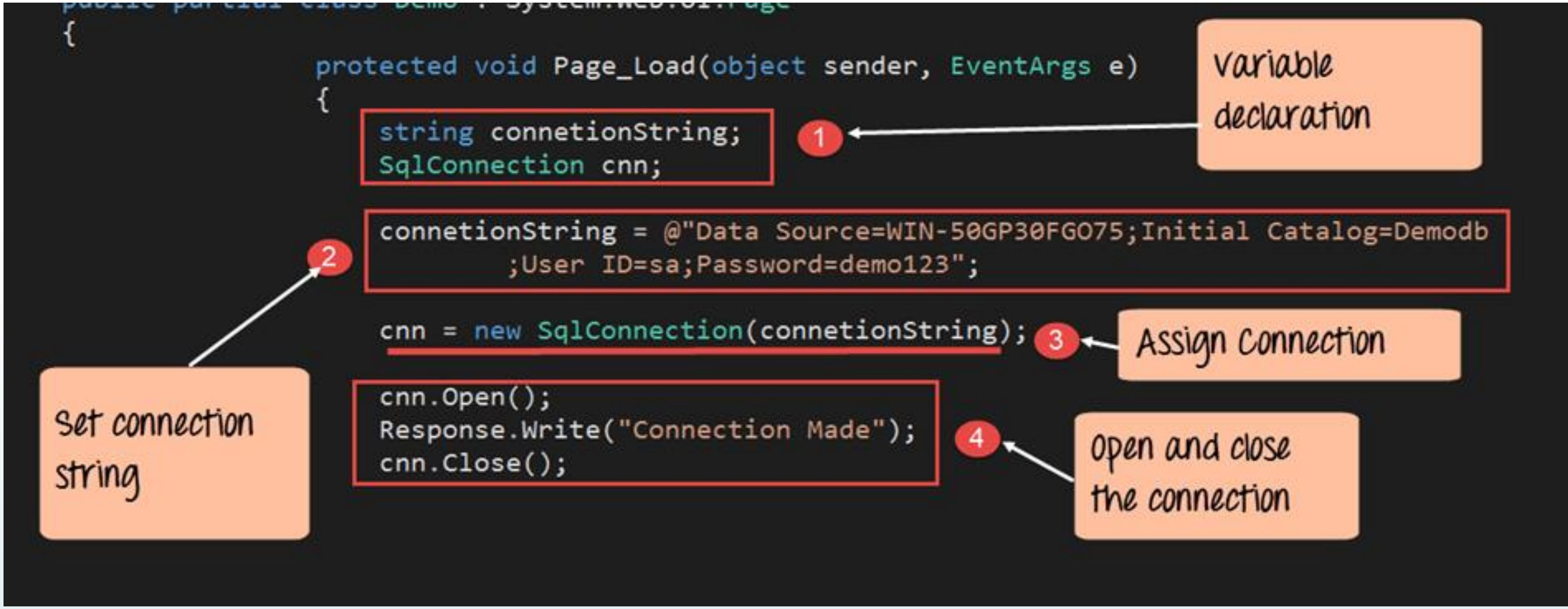
Something is done by the database but nothing is returned by the database.

Database connectivity

```
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data;
using System.Data.SqlClient;

namespace Employee
{

    public partial class login : System.Web.UI.Page
    {
        SqlConnection con = new SqlConnection("Data Source=(localdb)\\MSSQLLocalDB;Initial Catalog=employee;Integrated Security=T

        void dispdata()
        {
            SqlCommand cmd = con.CreateCommand();
            cmd.CommandType = CommandType.Text;
            cmd.CommandText = "select * from dbo.detail" ;
            cmd.ExecuteNonQuery();

            //****grid view binding without procedure
            DataTable dt = new DataTable();
            SqlDataAdapter da = new SqlDataAdapter(cmd);
            da.Fill(dt);
            GridView1.DataSource = dt;
            GridView1.DataBind();
        }


        protected void Page_Load(object sender, EventArgs e)
        {
            con.Open();
            dispdata();
        }
            //use of global variable usinf response.write
```

Copy the connection string

Command type is text for query



Example Copnnection string:

SqlConnection con = new SqlConnection("Data Source=(localdb)\\MSSQLLocalDB;Initial Catalog=employee;Integrated Security=True;Connect Timeout=30;Encrypt=False;TrustServerCertificate=False;ApplicationIntent=ReadWrite;MultiSubnet Failover=False");

SqlConnection con = new SqlConnection (“Data Source=192.168.52.71;Initial Catalog=Trainee_may19;User ID=TestDB;Password=TestDB”);

DataGird Connectivity

```
SqlConnection con = new SqlConnection("Data Source=(localdb)\\M

void dispdata()
{
    SqlCommand cmd = con.CreateCommand();
    cmd.CommandType = CommandType.Text;
    cmd.CommandText = "select * from dbo.detail" ;
    cmd.ExecuteNonQuery();

    //****grid view binding without procedure
    DataTable dt = new DataTable();
    SqlDataAdapter da = new SqlDataAdapter(cmd);
    da.Fill(dt);
    GridView1.DataSource = dt;
    GridView1.DataBind();
}


protected void Page_Load(object sender, EventArgs e)
{
    con.Open();
```

employee - Microsoft Visual Studio

File Edit View Project Build Debug Team Tools Test Analyze Window Help

Debug | Any CPU | IIS Express (Google Chrome)

login.aspx.cs* | Registration.aspx | Details.aspx | login.aspx

Employee | Employee.login | Button1_Click(object sender, EventArgs e)

```csharp

        protected void Button1_Click(object sender, EventArgs e)
        {
          SqlCommand cmd = con.CreateCommand();
          cmd.CommandType = CommandType.Text;
          cmd.CommandText = " insert into dbo.detail values('"+TextBox1.Text+"','"+TextBox2.Text+"','"+(Convert.ToInt32(Tex
          cmd.ExecuteNonQuery();
          dispdata();

        }
    //use of stored procedure with and without parameter

        protected void Button2_Click(object sender, EventArgs e)
        {
            SqlCommand cmd = con.CreateCommand();
            cmd.CommandType = CommandType.StoredProcedure;
            cmd.CommandText = "[dbo].[Getdetails]";
            cmd.Connection = con;

            GridView1.EmptyDataText = "No details available";
            GridView1.DataSource = cmd.ExecuteReader();
            GridView1.DataBind();
        }

        protected void Button3_Click(object sender, EventArgs e)
        {
            SqlCommand cmd = con.CreateCommand();
            cmd.CommandType = CommandType.StoredProcedure;
            cmd.CommandText = "[dbo].[Getdetail_id]";
            cmd.Parameters.Add("@Name", SqlDbType.VarChar).Value = TextBox1.Text.Trim();
            cmd.Connection = con;
            GridView1.EmptyDataText = "No details available";
            GridView1.DataSource = cmd.ExecuteReader();
            GridView1.DataBind();
```

100 %

Activate Windows
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Ready | Ln 43 | Col 9 | Ch 9 | INS | Add to Source C

Steps:

- The first step is to create variables. It will be used to create the connection string and the connection to the SQL Server database.
- The next step is to actually create the connection string. The connection string consists of the following parts
- Data Source – This is the name of the server on which the database resides..
- The Initial Catalog is used to specify the name of the database
- The UserID and Password are the credentials required to connect to the database.
- Next, we assign the connecting string to the variable ‘cnn’.
  - The variable cnn is of type SqlConnection. This is used to establish a connection to the database.
  - SqlConnection is a class in ASP.Net, which is used to create a connection to a database.
  - To use this class, you have to first create an object of this class. Hence, here we create a variable called ‘cnn’/ con which is of the type SqlConnection.
  - Next, we use the open method of the cnn variable to open a connection to the database. We display a message to the user that the connection is established. This is done via the ‘response.write’ method. We then close the connection to the database.

- Insert command:
-
```
cmd.CommandText = " insert into dbo.detail values('"+TextBox1.Text+"','"+TextBox2.Text+"','"+(Convert.ToInt32(TextBox3.Text))+"' )";
```

State Management

When a new instance of the Web page class is created each time the page is posted to the server. In traditional Web programming, this would typically mean that all information associated with the page and the controls on the page would be lost with each round trip. ASP.NET includes several options that help you preserve data on both a per-page basis and an application-wide basis.

- View state
- Control state
- Hidden fields
- Cookies
- Query strings

Client based options

- Application state
- Session state
- Profile Properties

Server based options

ANAND RATHI

behind every successful investor

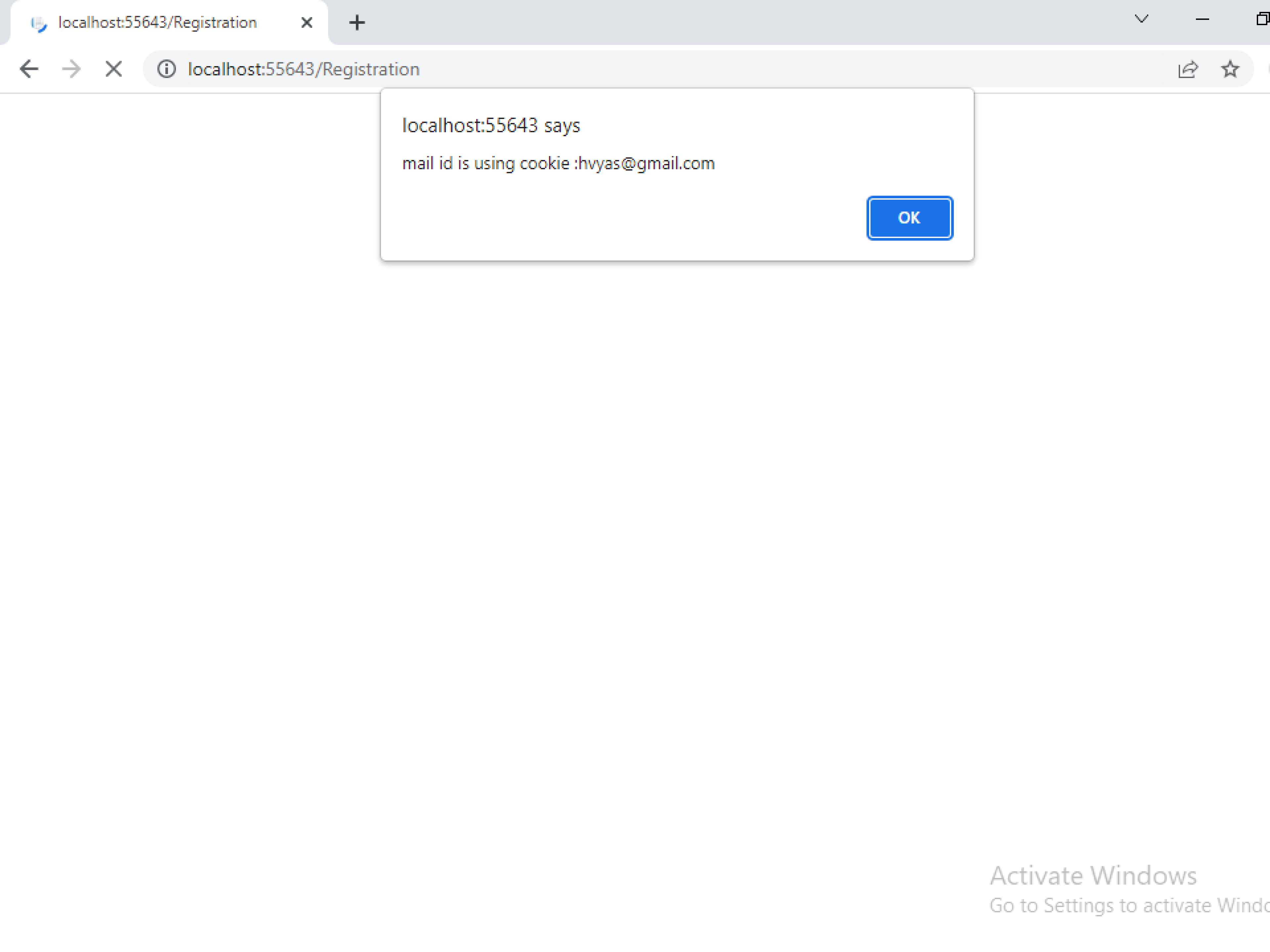
Cookies

- Cookies are a State Management Technique that can store the values of control after a post-back.
- Cookies can store user-specific Information on the Client's machine like when the user last visited your site.
- Cookies are also known by many names, such as HTTP Cookies, Browser Cookies, Web Cookies, Session Cookies and so on.
- Basically cookies are a small text file sent by the web server and saved by the Web Browser on the Client's Machine.

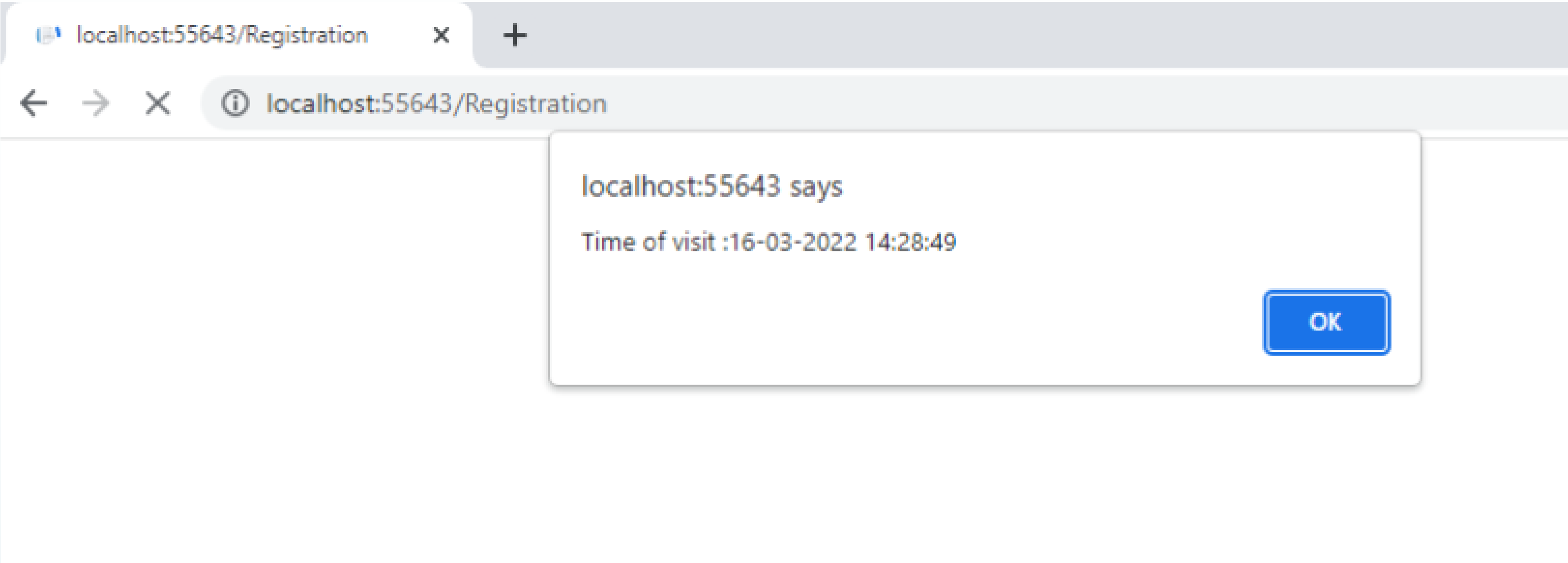
Basically Cookies are one of the following 2 types:

- **Persistent Cookies:** Persistent Cookies are Permanent Cookies stored as a text file in the hard disk of the computer.
- **Non-Persistent Cookies:** Non-Persistent cookies are temporary. They are also called in-memory cookies and session-based cookies. These cookies are active as long as the browser remains active, in other words if the browser is closed then the cookies automatically expire.

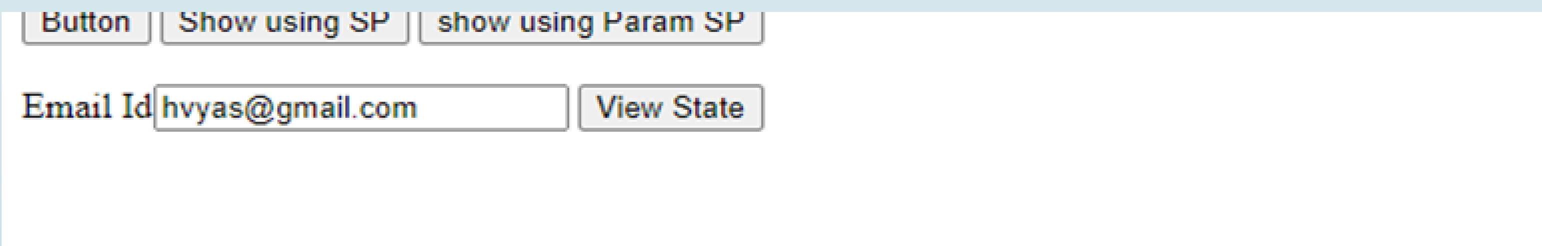
```
protected void Button1_Click(object sender, EventArgs e)
{   //Session["mailid"] = TextBox2.Text;
    //Response.Write("<script>alert('mail id is :" + TextBox2.Text + "');</script>");
    //// ***use of response.redirect
    // Response.redirect("login.aspx?id"+TextBox2.Text);

     //****use of cookie
    HttpCookie cID = new HttpCookie("cID");
    cID.Value = TextBox2.Text;
    Response.Cookies.Add(cID);
    Response.Write("<script>alert('mail id is using cookie :" + TextBox2.Text + "');</script>");
    //cID.Expires = DateTime.Now.AddMinutes(1);
    HttpCookie visit = new HttpCookie("visit");
    visit.Value = DateTime.Now.ToString();
    Response.Write("<script>alert('Time of visit :" + visit.Value + "');</script>");
}

protected void Button2_Click(object sender, EventArgs e)
{
    Response.Redirect("login.aspx");
}
```


localhost:55643/Registration
localhost:55643/Registration
localhost:55643 says
mail id is using cookie :hvyas@gmail.com
OK
Activate Windows
Go to Settings to activate Windo

cookies



In login page:



- Important points:

•From this output are the following 3 important points:

•If the same URL is open in a new tab in the same browser then it also provides the same output.

•Now when I close the browser and again open the browser and open the same URL then I also don't get the same output, in other words the **cookies expire that are the Non-Persistent Cookies.**

•For **Persistent Cookies we need to add the Expiration time** of the cookies, in other words the browser manages the time of the cookies and if we close the browser then we also get the same output and the cookie will not expire; they are Persistent Cookies.

Session State

In registration form

```
        }

        protected void Button1_Click(object sender, EventArgs e)
        {

            Session["mailid"] = TextBox2.Text;
            Response.Write("<script>alert('mail id is :" + TextBox2.Text + "');</script>");
            // ***use of response.redirect
            Response.Redirect("login.aspx?id" + TextBox2.Text);



        protected void Button2_Click(object sender, EventArgs e)
        {
            Response.Redirect("login.aspx");
        }
    }
```

login.aspx | Registration.aspx.cs | Registration.aspx* | login.aspx.cs* | Details.aspx | Details.aspx.cs | dbo.Procedure_1.sql

Employee | Employee.login | Button4_Click(object sender, EventArgs e)

```
            cmd.Connection = con;

            GridView1.EmptyDataText = "No details available";
            GridView1.DataSource = cmd.ExecuteReader();
            GridView1.DataBind();
        }

        protected void Button4_Click(object sender, EventArgs e)
        {
            if (Session["mailid"] != null)
                TextBox4.Text = Session["mailID"].ToString();

            //if (Request.Cookies["cID"] != null)
            //    TextBox4.Text = Request.Cookies["cID"].Value;

```

- **The advantages of using session state are as follows:**

  - It is easy to implement.
  - It ensures data durability, since session state retains data even if ASP.NET work process restarts as data in Session State is stored in other process space.
  - It works in the multi-process configuration, thus ensures platform scalability.

  The disadvantages of using session state are:

  - Since data in session state is stored in server memory, it is not advisable to use session state when working with large sum of data. Session state variable stays in memory until you destroy it, so too many variables in the memory effect performan

Application state

- **Disadvantages**
- Application State requires server resources to store data. This can lead to scalability issues if not handled properly.
- Application State is not thread safe, so we need to implement Locks.
- In the case of an application failure or restart, all the stored data is lost.

- Advantages of using cookies
- Here are some of the advantages of using cookies to store session state.
  · Cookies are simple to use and implement.
  · Occupies less memory, do not require any server resources and are stored on the user's computer so no extra burden on server.
  · We can configure cookies to expire when the browser session ends (session cookies) or they can exist for a specified length of time on the client's computer (persistent cookies).
  · Cookies persist a much longer period of time than Session state.
-
- Disadvantages of using cookies
- Here are some of the disadvantages:
- · As mentioned previously, cookies are not secure as they are stored in clear text they may pose a possible security risk as anyone can open and tamper with cookies. You can manually encrypt and decrypt cookies, but it requires extra coding and can affect application performance because of the time that is required for encryption and decryption
  · Several limitations exist on the size of the cookie text(4kb in general), number of cookies(20 per site in general), et

Thanks